Interpreting ViT Attention Maps Through a Fast-Weight Lens

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Abstract

Attention maps are often read as input importance, yet attention weights describe interactions between internal tokens. We develop an analytical framework based on the established fast-weight view, representing each attention head as an image-conditioned two-layer network. The framework separates three reading assumptions: image specificity, spatial correspondence and output contribution. Each is examined through a diagnostic proxy, a correction and a matched control, with the backbone fixed. On DeiT and DINO, norm weighting improves localisation and human alignment in supervised DeiT, whereas index-based relocation degrades map quality. Norm weighting's deletion-score gains remain below the registered practical margin. The contribution is a structured way to examine how internal attention computations are translated into spatial explanations.

1 Introduction

Attention maps are read as input importance in vision transformers (ViTs) [8, 9, 13], yet their coefficients describe interactions between contextualised tokens. Translating those interactions into image evidence requires additional assumptions. High-norm background artefacts [11, 18] highlight this interpretive gap.

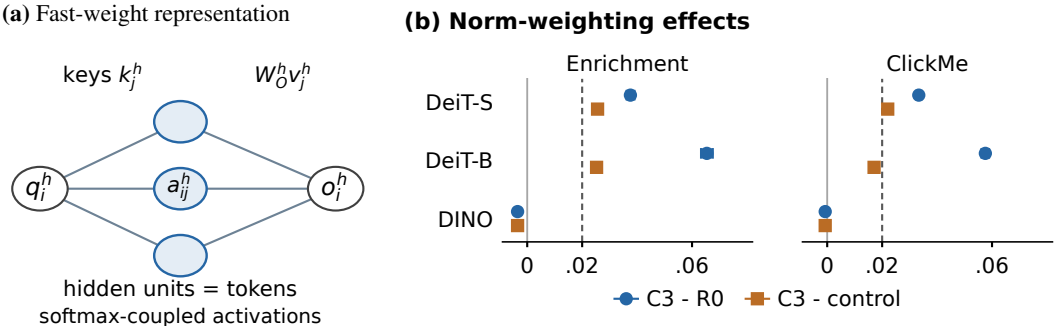


Figure 1: (a) Keys and transformed values are image-conditioned weights; tokens are softmax-coupled units. (b) Norm weighting (C3) minus received attention (R0) or zero-high-norm control, layers 9–12 (DeiT-S/B, DINO). Points/bars: paired means/95% image-bootstrap intervals. Dashed: registered 0.02 margin.

We analyse *received attention* (summed over queries/heads at each patch) through the fast-weight lens in Fig. 1a.

Preprint.

(1) **An analytical framework** separating three assumptions behind the translation from internal attention to spatial importance.
(2) **Controlled tests** linking each assumption to a diagnostic proxy, a correction and a matched control on frozen DeiT-S/B and DINO ViT-S.
(3) **Bounded empirical insights:** useful alignment gains from norm weighting, failed index-based relocation, and limited benefit from combining corrections.

2 Related work

2.1 Fast-weight foundations. Fast-weight memories use one network to write another's temporary weights [26]. Linear attention admits a fast-weight-programming interpretation [25], and softmax attention connects to modern Hopfield memory [24]. MiTA treats attention as an input-conditioned two-layer network to organise efficient attention [29]. We build on this representation for a different purpose: making attention-map reading assumptions explicit, then examining their proposed corrections.

2.2 Attention readings and token correspondence. Attention's explanatory role remains debated [17, 30]. Rollout [1], relevance propagation [9], gradients [10] and filtering [3] change map construction. Norm-based analysis [19, 33], like CAM's output weighting [32], distinguishes activation from contribution. GlobEnc/ALTI include further components and cross-layer aggregation [15, 22]; our score isolates local attention terms. Contextual mixing need not erase token-identity or input-location information [7, 23]; slot decodability does not establish content provenance. Registers alter the model [11, 18], with no-op heads providing a functional analogy [6]; we fix the backbone.

2.3 Evaluating explanations. Visually plausible maps may ignore learned parameters [2], supervision changes ViT maps [28], and deletion depends on occlusion [5]. CNN centre-bias findings [20] motivate spatial controls without establishing the same mechanism in ViTs. We therefore separate localisation/human alignment from deletion behaviour, and compare each correction with a control for a specific alternative explanation.

3 Methodology: A Fast-Weight Analysis Framework

3.1 From internal computation to a spatial map. Normalised tokens $x_1, \dots, x_{N+1}$ (CLS and $N = 196$ patches) enter H attention heads. Affine projections form d-dimensional queries/keys and values (App. A); before residual addition,

$$o_i = b_O + \sum_{h,j} a_{ij}^h W_O^h v_j^h, \qquad a_{ij}^h = \mathrm{softmax}_j\big(q_i^{h\top} k_j^h / \sqrt{d}\big). \tag{1}$$

For a fixed image, each head is a two-layer network in q_i^h, with input weights $k_j^h/\sqrt{d}$, coupled softmax activations a_{ij}^h and output weights $W_O^h v_j^h$. Dropout is inactive at evaluation. This exact representation separates unit activation from output contribution; it does not locate a unit's input evidence. Received attention $s_j = \sum_{h,i} a_{ij}^h$ places activation at patch j. Queries include all tokens; maps exclude CLS and sum layers 9–12, fixed before formal runs.

3.2 Three assumptions and their controlled tests. Unit meaning requires measurement [4]. We operationalise three questions through proxies (Fig. 2a–c) and corresponding corrections (d–f).

Image specificity: does the map reflect this image rather than shared structure? We measure mean-map variance share and subtract the held-out mean (C1: $s - \bar{s}$), against centred-Gaussian subtraction.

Spatial correspondence: does a token slot identify its input evidence? We measure index decodability and test $r_p = \sum_j s_j P(p \mid x_j)$ (C2), against blur matched in mean squared token displacement. The probe decodes indices, not provenance.

Output contribution: does activation alone explain a token's output term? We measure high-norm attention and test $c_j = \sum_{h,i} a_{ij}^h \|W_O^h v_j^h\|_2$ (C3) against zeroing high-norm patches. This norm-weighted component [19, 33] omits cancellation and later computation. Benefit requires gains over R0 *and* control; calibration: App. A.

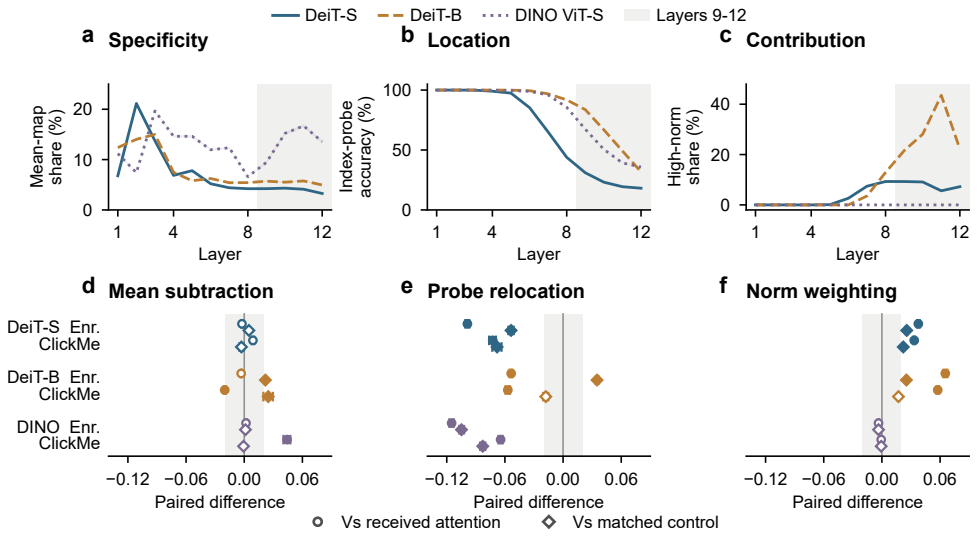


Figure 2: Diagnostic proxies (a–c) and correction effects (d–f), paired by column. Top: per-layer means/95% intervals; grey: layers 9–12. Bottom: correction minus comparator, layers 9–12 (Enr.: enrichment). Circles: R0; diamonds: centred-Gaussian subtraction (d), displacement-matched blur (e), zero-high-norm control (f). Bars: 95% paired image-bootstrap intervals, unadjusted. Filled/open: higher or lower/comparable. Grey bands: ± 0.02 margin, not equivalence. Diagnosis–benefit interpretation is post hoc; samples: App. A.

3.3 Evaluation and decision rules. A centred Gaussian correlates 0.41 with raw ClickMe maps. Our *centre-controlled* metrics are *same-ring enrichment*, selected-patch object rate minus the rate at the same radius, using ImageNet-S masks [16]; and *image-specific alignment*, own-image minus other-image Spearman correlation with ClickMe maps [14, 21]. The *deletion score* adapts LIF–MIF area difference [5] to top-1 preservation under token removal. With 1,000 paired image-bootstrap resamples, a difference is *higher/lower* if its 95% interval excludes zero and its magnitude reaches 0.02 (0.05 for deletion); otherwise it is *comparable*, a registered category, not equivalence.

4 Experiments

4.1 Setup and registered predictions. Frozen DeiT-S/B [27] and DINO ViT-S/16 [8] are tested on ImageNet [12], ImageNet-S (12,419 images) and ClickMe (17,408). A separate 4,000-image set splits equally into probe training and calibration/test. Size-S references use a register [18] and released MiTA checkpoint [29]. **R1:** DINO norm weighting and relocation at layers 1–4 remain comparable. **R2:** the combination improves all four supervised alignment comparisons. **R3:** neither model-side variant is higher than DeiT-S. Combination–R0 and correction–control contrasts are primary; other correction–R0 contrasts are descriptive except R1. Diagnosis–benefit interpretation is post hoc. Registration and prior observations: App. A.

4.2 What the diagnostics show (Fig. 2a–c). Shared mean-map structure is largely radial but accounts for limited variance; patch-index accuracy declines with depth; high-norm attention concentrates in supervised models and is negligible in DINO. These are different properties of internal computation, not interchangeable indicators of map quality. Per-layer values, execution tests and parameter sensitivity appear in App. B.

4.3 Which corrections help (Table 1; Fig. 2d–f). Contribution magnitudes improve alignment in a limited setting. C3 raises enrichment by $0.038/0.065$ and ClickMe by $0.033/0.058$ in DeiT-S/B. Three comparisons also beat its control; DeiT-B ClickMe is comparable. Deletion gains ($0.017/0.029$) stay below 0.05, and DINO changes are at most 0.004 (R1). These results support a useful norm-based reading within the tested settings.

Table 1: Layers 9–12; higher is better. Arrows: higher/lower versus R0 (DeiT-S for model-side rows); unmarked tested cells: comparable. $\diamond$: higher than control; **bold**: both comparisons. Indented rows: controls. GAE [10]: descriptive, gradients, all 12 layers. Enr./Click./Del.: enrichment/ClickMe/deletion. Paired intervals: Table B.2; tests use unrounded data.

	DeiT-S			DeiT-B			DINO ViT-S		
Reading	Enr.	Click.	Del.	Enr.	Click.	Del.	Enr.	Click.	Del.
Received attention	0.140	0.157	0.236	0.080	0.107	0.169	0.199	0.208	0.285
Contribution weighting	**0.178**$^{\uparrow\diamond}$	**0.190**$^{\uparrow\diamond}$	0.253	**0.146**$^{\uparrow\diamond}$	0.165$^{\uparrow}$	0.198	0.196	0.207	0.285
zero high-norm tokens	0.152	0.168	0.241	0.121$^{\uparrow}$	0.148$^{\uparrow}$	0.186	0.199	0.208	0.285
Mean-map subtraction	0.138	0.165	0.208	0.077$^{\diamond}$	0.087$^{\downarrow\diamond}$	0.144$^{\diamond}$	0.201	0.252$^{\uparrow}$	0.233$^{\downarrow}$
subtract centre Gaussian	0.133	0.168	0.167$^{\downarrow}$	0.056$^{\downarrow}$	0.062$^{\downarrow}$	0.024$^{\downarrow}$	0.200	0.253$^{\uparrow}$	0.232$^{\downarrow}$
Probe relocation	0.041$^{\downarrow}$	0.084$^{\downarrow}$	0.076$^{\downarrow}$	0.027$^{\downarrow\diamond}$	0.050$^{\downarrow}$	0.051$^{\downarrow}$	0.084$^{\downarrow}$	0.143$^{\downarrow}$	0.141$^{\downarrow}$
matched blur	0.095$^{\downarrow}$	0.152	0.181$^{\downarrow}$	−0.008$^{\downarrow}$	0.068$^{\downarrow}$	0.097$^{\downarrow}$	0.189	0.226	0.287
All three corrections	0.059$^{\downarrow}$	0.123$^{\downarrow}$	0.076$^{\downarrow}$	0.079	0.137$^{\uparrow}$	0.085$^{\downarrow}$	0.088$^{\downarrow}$	0.177$^{\downarrow}$	0.090$^{\downarrow}$
+ test-time register	0.152	0.169	0.236	–	–	–	–	–	–
Top-k training (MiTA)	0.100$^{\downarrow}$	0.115$^{\downarrow}$	–	–	–	–	–	–	–
GAE (reference)	0.204	0.234	0.345	0.196	0.213	0.308	0.221	0.234	0.418

Other proxy-guided changes lack consistent benefits. Mean subtraction's only R0 gain (DINO ClickMe) is comparable to its control. Relocation lowers all nine deep-layer comparisons and underperforms matched blur in DeiT-S/DINO. Early relocation remains comparable (R1). R2 *partly holds*: one supervised comparison is higher, one comparable and two lower.

Model-side references have narrower scope. Register gains ($0.012/0.013$) stay below 0.02; MiTA scores lower but differs in training/preprocessing. R3 holds within these comparisons, without establishing general superiority of reading changes.

4.4 What the tests establish. Assigned weights outperform shuffled weights, but weak dose–gain associations (Spearman 0.045–0.158; App. B) do not establish diagnostic prediction. Failed relocation cautions against treating index decodability as content provenance; its cause remains unresolved after local-error/MLP checks (App. C). Useful correction and validated interpretation remain separate claims.

5 Discussion

5.1 Scope and limitations. The framework separates correction benefit from interpretation validity. Evidence covers three frozen S/B classifiers and received attention; CLS maps, larger/register-trained models, repeated training seeds and threshold sensitivity are untested. ClickMe may overlap model training data. Improvements are visually subtle (App. D) and below GAE's descriptive scores. Bootstrap intervals condition on fixed checkpoints, probes and pairing. General diagnostic prediction remains open, requiring stronger token-to-input attribution and broader model coverage.

5.2 Ethical considerations. Public images and checkpoints contain people and known biases. Better alignment offers partial evidence about model behaviour; these maps should not be treated as complete explanations.

6 Conclusion

Our fast-weight framework connects reading assumptions to controlled comparisons. Norm weighting benefits DeiT alignment, index-based relocation fails, and the registered combination lacks consistent gains. Alignment and deletion behaviour require separate evaluation.

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A Experimental details

Models. All models are frozen ViTs with 16×16 patches on 224×224 inputs, taken from timm [31]. *DeiT-S and DeiT-B* [27]: `deit_small_patch16_224.fb_in1k` and `deit_base_patch16_224.fb_in1k` (resize to 248, centre crop 224: a crop ratio of 0.9). *DINO ViT-S/16* [8]: `vit_small_patch16_224.dino` with the official DINO linear head (resize to 256, centre crop 224: a crop ratio of 0.875). *DeiT-S with a test-time register* [18]: the activations of eight feed-forward units in layers 4 and 5 are moved to one added register token, which is left out of every map, like CLS. The units were selected on 500 ImageNet-S images, where they reduce the number of high-norm patch tokens (counted per layer) by 92.75% and keep 99.0% of the predictions; on the other 11,919 images the reduction is 91.5%, and ImageNet accuracy stays at 79.8%. For DeiT-B the same search found no such units (44.9% reduction with 32 units), so we report the register for DeiT-S only. *Top-k training*: the DeiT-S checkpoint released with MiTA [29] (25 summary slots, $k = 25$), with its own preprocessing (crop ratio 0.875). It differs from our DeiT-S in training and preprocessing, so its row in Table 1 is not a controlled comparison; its deletion score is undefined because its summary slots pool fixed regions of the full grid. Both model-side fixes are tested at size S only; in a preliminary comparison with an earlier protocol, top-k training scored 0.011 lower in object enrichment than its baseline at size S but 0.037 higher at size Tiny, so the result should not be read beyond DeiT-S.

Attention conventions. With x the block input after its first layer norm, projections include the checkpoint's biases:

$$q_i^h = W_Q^h x_i + b_Q^h, \quad k_j^h = W_K^h x_j + b_K^h, \quad v_j^h = W_V^h x_j + b_V^h.$$

Equation 1 includes the output bias b_O and describes the attention branch before residual addition, with dropout inactive at evaluation. W_O^h is the head's slice of the output projection. C3 assigns the per-head vector magnitudes to patches; the shared output bias is not assigned to any patch. The fixed-image two-layer representation uses the coupled softmax activation, rather than an elementwise nonlinearity.

Data. *Execution test*: 5,000 randomly drawn ImageNet validation images [12]. *ImageNet-S* [16]: all 12,419 validation images with segmentation masks; enrichment is defined for 12,150 (12,144 for DINO) after excluding images without valid foreground or background patches. Deletion uses a fixed random subset of 2,000. *ClickMe* [14, 21]: the 17,408-image validation split released with Harmonization (17 `val-*.tfrecords` shards). This split name does not identify the original ImageNet split of each image. Images are resized from 256×256 to 224×224 without cropping; heatmaps are resized to 64×64, blurred with a 10×10 Gaussian ($\sigma = 10$), then resized to 224×224 (bilinear interpolation). *Held-out set*: 4,000 ImageNet validation images excluded from map-quality and execution evaluation; 2,000 train probes and 2,000 provide mean maps, probe-test accuracies and blur calibration. Our near-duplicate check matched 2 of 4,002 candidate ImageNet validation images to ClickMe records; this does not establish every ClickMe image's training-set membership. Possible overlap with model training images limits generalisation claims.

Diagnostics. All three are computed per layer. *Specificity*: with $\bar{s}$ the mean received-attention map of the held-out images and Var_j the variance over the 196 patches of one image, the share explained is $1 - \sum_x \mathrm{Var}_j(s - \bar{s}) \,/\, \sum_x \mathrm{Var}_j(s)$, summed over the ImageNet-S images x. *Location*: the top-1 accuracy of the linear position probe on the 196 patch tokens of each of 2,000 held-out images. *Contribution*: the share of received attention on high-norm tokens, on the ImageNet-S images. The high-norm threshold (three times the median, defined below) is an operational definition fixed before the experiments; we did not test other thresholds. The correction itself uses no threshold.

Corrections and matched controls. Scores are computed per layer and summed over layers 9–12 without per-layer rescaling, except for the C1 control's calibration described below. Queries include all 197 tokens; maps contain 196 patches. Negative C1 scores are retained for ranking, without clipping. *Mean-map subtraction* computes $\sum_\ell (s_\ell - \bar{s}_\ell)$ using the 2,000 calibration images. With $S = \sum_\ell s_\ell$ and $\bar{S} = \sum_\ell \bar{s}_\ell$, its control is $S - G\ \mathrm{sd}_j(\bar{S}) / \mathrm{sd}_j(G)$, where G is a centred isotropic Gaussian ($\sigma = 3.5$ patches). Calibration thus uses the *summed* mean map. *Probe relocation* computes $r_p = \sum_j s_j P_\ell(p \mid x_j)$, preserving total score. One linear 196-way classifier per model/layer is

trained on standardised tokens after the first layer norm: cross-entropy, Adam, learning rate 0.01, 30 epochs, token minibatches of 4,096, seed 0. Feature means/standard deviations use only the 2,000 training images; standard deviations are floored at 10^{-6}. Its Gaussian control is a row-normalised kernel $K_{jp} \propto \exp(-\|p-j\|^2/(2\sigma_\ell^2))$. The width (0.7–2.9 patches) matches

$$D_\ell = \frac{1}{2000N} \sum_{x,j,p} P_\ell(p \mid x_j)\|p-j\|^2$$

on calibration images, using uniform averaging over source patch indices. It does not match attention-weighted score-mass displacement. *Contribution weighting* computes $c_j = \sum_{h,i} a_{ij}^h \|W_O^h v_j^h\|_2$, a sum of per-head norm-weighted terms related to Kobayashi et al. [19]. Summing magnitudes before aggregating across heads omits cross-head cancellation; residual addition, subsequent normalisation and later-layer computation are outside this score. The control zeros received attention at high-norm patches: block-input norm greater than three times the image's median patch norm, using the midpoint median. C3 itself has no high-norm threshold. *All three corrections*: contribution weighting, probe relocation, then subtraction of the held-out mean of the resulting map, per layer before summation.

Metrics. *Same-ring object enrichment.* A patch is valid if it contains at least one non-ignored pixel, and is an object patch if at least half of its non-ignored pixels belong to the image's class mask. We select as many top-scoring patches as there are object patches. For every ring of patches at the same distance from the grid centre, we take the object rate among its selected patches minus that among all its patches, and average over rings, weighted by their selected patches. Invalid patches are excluded from selection and ring baselines. Ties are broken with the fixed seeded perturbation used in the original runs. A purely radial map scores zero in expectation over tie-breaking. *Image-specific ClickMe alignment.* The map is upsampled to 224×224 and correlated (Spearman, over pixels) with the image's own human map; we subtract its correlation with the previous image's human map within each batch (`roll(1)`; batch size 64). For an image-independent map, these differences average to zero over the paired set because the control permutes the same human maps; individual-image differences need not be zero. This removes fixed-prior alignment on average, without guaranteeing invariance to a prior mixed into an image-dependent map. *Deletion score.* Patches are ranked once by the original map. At removal budgets $0, 1, 2, 4, 8, 16, 32, 64, 98, 147$, we record the binary indicator that the original top-1 prediction is preserved. Tokens are dropped after positional embedding; CLS and any register remain. We use an SRG-style LIF–MIF area difference [5], adapting the value function from class probability to top-1 preservation. Areas use trapezoidal integration over removed-patch fraction $b/196$, from 0 to $147/196$. *Validity of the metrics.* A random map, a centred Gaussian and the mean map all score within 0.004 of zero on enrichment and on ClickMe alignment (the two image-independent maps have exactly zero mean on the latter, up to numerical precision); on deletion the centred Gaussian scores 0.110, 0.099 and 0.140 (DeiT-S, DeiT-B, DINO ViT-S), which is why every correction is compared with a matched control. *Statistics.* One value per image; paired image-bootstrap with 1,000 resamples, seed 0 and 95% percentile intervals. A difference is *higher* (*h*) or *lower* (*l*) if its interval excludes zero and its absolute mean reaches 0.02 (0.05 for deletion), and *comparable* (*c*) otherwise. These unchanged registered categories do not establish equivalence. Intervals are unadjusted for multiple comparisons. Resampling holds checkpoints, probes and the within-batch ClickMe pairing fixed, excluding uncertainty from training, probe fitting or alternative pairings. Because the control reuses human maps within each batch, ClickMe score rows are not independent pairings.

GAE reference. The gradient-based reference [10] targets the unmodified model's predicted top-1 logit and uses all 12 attention layers. Starting with $R = I$, each layer updates $R \leftarrow R + \mathrm{mean}_h[(\nabla_{A^h} f_c \odot A^h)_+]R$, then takes the final CLS-to-patch row. It is class-specific and spans all layers, whereas R0/C1–C3 are class-agnostic and sum layers 9–12. Its scores provide descriptive context rather than a matched single-component ablation.

Registered predictions. The protocol was frozen before formal runs, after smoke tests on 32–64 images and earlier exploratory observations documented in `docs/preregistration.md`. Historical P4/P3/P6 map to R1/R2/R3; definitions, margins and verdicts are unchanged. R2 has three possible outcomes: holds (4/4 higher), partly holds (1–3) and does not hold (0). Primary contrasts were the combination versus R0 and each correction versus its matched control, without a registered

direction for the latter. Other correction–R0 contrasts are descriptive, except R1's registered negative controls. Prior execution/diagnostic results had been seen, making the last two rows of Table A.1 replications. Diagnosis–benefit interpretation and supplementary mechanism analyses are post hoc.

Table A.1: Registered predictions and outcomes (centre-controlled metrics: enrichment and ClickMe alignment).

	Prediction	Outcome
R1	Contribution weighting is comparable to received attention in DINO ViT-S on all three metrics	holds (3 of 3 cells)
R1	Relocation at layers 1–4 is comparable to received attention on the centre-controlled metrics in all models	holds (6 of 6 cells)
R2	All three corrections together are higher than received attention on the centre-controlled metrics in DeiT-S and DeiT-B	partly holds (1 higher: DeiT-B ClickMe; 1 comparable; 2 lower)
R3	Neither model-side change is higher than DeiT-S received attention on the centre-controlled metrics	holds (4 of 4 cells)
–	Execution agreement rises with k and exceeds that of random sets at every k	holds in all models
–	Probe accuracy falls with depth; the share of attention on high-norm tokens peaks in layers 6–12 of the supervised models and is below 1% in every layer of DINO ViT-S	holds in all models

Compute and code. Experiments took about 19 NVIDIA L4 GPU hours. Code, configurations, fixed splits, checkpoint revisions and recorded provenance: `https://github.com/hhuang999/vit-fastweight-lens`. The release supplies `configs/`, `data/splits/` and `docs/provenance/`; unavailable historical environment fields remain unknown. Presentation changes use stored evidence, without rerunning experiments.

B Additional results

These results support the framework at different levels: execution and parameter sensitivity check internal dependence, while paired comparisons and supplementary analyses assess map quality and proposed interpretations.

Execution test. The 14×14 grid is pooled into 16 overlapping regions. In every layer/head, each region's mean query selects its top-k keys; CLS retains its own set. Sets from an unmodified forward pass restrict every query to its region's set in all 12 layers simultaneously. Top-k sets beat equal-size random sets at every tested k (4–128) in all models, and agreement rises with k (Fig. B.1). The dotted attention-removed baseline alone uses uniform crop ratio 0.875; solid/dashed DeiT curves use 0.9.

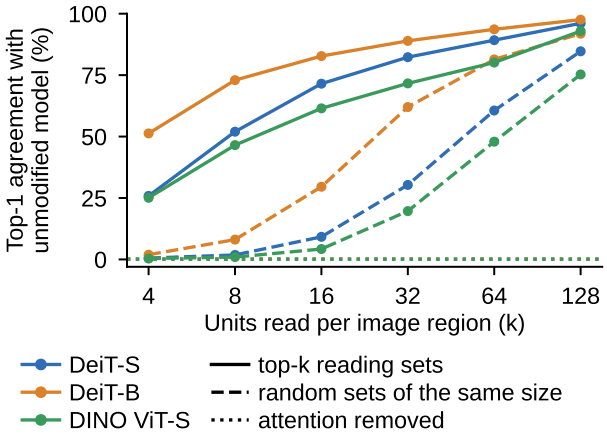


Figure B.1: Top-1 agreement on 5,000 ImageNet images with region-selected top-k units (solid), random units (dashed) or attention removed (dotted). Stored 95% intervals are at most ± 1.5 percentage points. Dotted curves use crop ratio 0.875 for all models; other curves use each checkpoint's main preprocessing.

Diagnostics per layer. Table B.1 lists the diagnostics plotted in Fig. 2a–c and the uncorrected per-layer ClickMe alignment. Two further details complement the table. First, in layers 9–12, 82–98% of the variance of the mean map is explained by the distance from the grid centre. Second, averaged over layers 9–12, the mean value-vector norm $\|v_j\|$ of high-norm tokens is 0.29 (DeiT-S) and 0.23 (DeiT-B) times the median of the other patches (measured on 11,919 ImageNet-S images using uniform crop ratio 0.875, unlike the main DeiT evaluation at 0.9).

Recovery at the last layer. In the supervised models, single-layer received attention is worse at layer 12 than at the best of layers 8–11 (ClickMe values in Table B.1). Applied to layer 12, contribution weighting recovers 62% and 33% of this drop on enrichment (DeiT-S, DeiT-B), 31% and 45% on ClickMe alignment, and 8% and 29% on deletion. The other two corrections make layer 12 worse.

Table B.1: Per-layer diagnostics (Fig. 2a–c) and uncorrected ClickMe alignment (last three columns). S, B, D: DeiT-S, DeiT-B, DINO ViT-S; probe chance: 0.5%. Values are rounded; DINO layers 9–12 have exactly zero high-norm attention on this dataset.

	Specificity: mean-map share (%)			Location: probe accuracy (%)			Contribution: high-norm share (%)			Map quality: ClickMe alignment		
Layer	S	B	D	S	B	D	S	B	D	S	B	D
1	6.8	12.4	11.2	100.0	100.0	99.9	0.0	0.0	0.0	−0.108	−0.111	−0.089
2	21.2	14.0	7.4	100.0	100.0	100.0	0.0	0.0	0.0	0.007	0.041	0.064
3	13.7	15.0	19.7	99.9	100.0	99.9	0.0	0.0	0.0	0.081	0.004	0.056
4	6.9	7.5	14.6	99.0	100.0	99.8	0.0	0.0	0.0	0.132	0.110	0.140
5	7.8	5.8	14.6	97.5	99.9	99.6	0.1	0.0	0.0	0.148	0.130	0.150
6	5.2	6.2	11.9	85.3	99.5	98.9	2.6	0.0	0.0	0.136	0.151	0.167
7	4.4	5.4	12.4	65.0	96.8	95.9	7.4	3.5	0.0	0.147	0.131	0.150
8	4.2	5.4	6.6	43.9	91.8	85.6	9.3	12.9	0.0	0.149	0.118	0.170
9	4.2	5.7	9.5	31.0	83.6	67.4	9.3	21.5	0.0	0.184	0.112	0.198
10	4.3	5.5	15.2	23.1	66.8	50.5	9.1	28.1	0.0	0.171	0.139	0.207
11	4.1	5.8	16.7	19.4	49.2	39.2	5.6	43.5	0.0	0.149	0.120	0.211
12	3.3	5.0	13.6	18.1	31.8	35.7	7.2	22.2	0.0	0.098	0.023	0.179

Differences behind Table 1. Table B.2 gives the paired differences with their intervals. The values in Table 1 have 95% intervals of at most ± 0.005 on enrichment and ClickMe alignment and ± 0.012 on deletion. The model-side differences from DeiT-S received attention are $+0.012$ [0.011, 0.013] on enrichment, $+0.013$ [0.012, 0.013] on ClickMe and 0.000 [-0.005, 0.004] on deletion for the

Table B.2: Paired differences, layers 9–12: mean, 95% image-bootstrap interval and verdict (*h*/*c*/*l*: higher/comparable/lower). Margins: 0.02 (enrichment/ClickMe), 0.05 (deletion); intervals unadjusted. Samples: App. A. Top: descriptive correction–R0 contrasts and the primary combination–R0 contrasts. Bottom: primary correction–control contrasts. Registered negative controls are also discussed in App. A.

Model	Same-ring enrichment	ClickMe alignment	Deletion score
Contribution weighting − received attention			
DeiT-S	0.038 [0.036, 0.039] *h*	0.033 [0.032, 0.034] *h*	0.017 [0.013, 0.021] *c*
DeiT-B	0.065 [0.063, 0.067] *h*	0.058 [0.056, 0.059] *h*	0.029 [0.024, 0.034] *c*
DINO ViT-S	−0.004 [−0.004, −0.003] *c*	−0.001 [−0.001, −0.001] *c*	0.000 [−0.003, 0.004] *c*
Mean-map subtraction − received attention			
DeiT-S	−0.002 [−0.003, −0.002] *c*	0.009 [0.008, 0.010] *c*	−0.028 [−0.034, −0.023] *c*
DeiT-B	−0.003 [−0.004, −0.003] *c*	−0.020 [−0.021, −0.019] *l*	−0.025 [−0.031, −0.019] *c*
DINO ViT-S	0.002 [0.001, 0.003] *c*	0.044 [0.040, 0.048] *h*	−0.052 [−0.059, −0.044] *l*
Probe relocation − received attention			
DeiT-S	−0.099 [−0.101, −0.097] *l*	−0.073 [−0.076, −0.069] *l*	−0.161 [−0.171, −0.150] *l*
DeiT-B	−0.054 [−0.055, −0.052] *l*	−0.057 [−0.059, −0.055] *l*	−0.118 [−0.127, −0.110] *l*
DINO ViT-S	−0.115 [−0.117, −0.113] *l*	−0.064 [−0.067, −0.062] *l*	−0.144 [−0.153, −0.134] *l*
All three corrections − received attention			
DeiT-S	−0.081 [−0.083, −0.079] *l*	−0.034 [−0.038, −0.030] *l*	−0.160 [−0.171, −0.149] *l*
DeiT-B	−0.001 [−0.003, 0.001] *c*	0.030 [0.027, 0.032] *h*	−0.084 [−0.093, −0.076] *l*
DINO ViT-S	−0.111 [−0.114, −0.109] *l*	−0.031 [−0.035, −0.027] *l*	−0.195 [−0.205, −0.184] *l*
Contribution weighting − zero high-norm tokens (its control)			
DeiT-S	0.026 [0.025, 0.027] *h*	0.022 [0.021, 0.023] *h*	0.012 [0.008, 0.016] *c*
DeiT-B	0.025 [0.024, 0.026] *h*	0.017 [0.016, 0.018] *c*	0.013 [0.008, 0.018] *c*
DINO ViT-S	−0.004 [−0.004, −0.003] *c*	−0.001 [−0.001, −0.001] *c*	0.000 [−0.003, 0.004] *c*
Mean-map subtraction − centre-Gaussian subtraction (its control)			
DeiT-S	0.005 [0.004, 0.005] *c*	−0.003 [−0.005, −0.001] *c*	0.040 [0.034, 0.047] *c*
DeiT-B	0.022 [0.021, 0.023] *h*	0.025 [0.020, 0.030] *h*	0.120 [0.109, 0.130] *h*
DINO ViT-S	0.001 [0.001, 0.002] *c*	−0.001 [−0.001, 0.000] *c*	0.001 [−0.003, 0.005] *c*
Probe relocation − matched blur (its control)			
DeiT-S	−0.054 [−0.056, −0.051] *l*	−0.068 [−0.073, −0.064] *l*	−0.105 [−0.116, −0.095] *l*
DeiT-B	0.035 [0.034, 0.036] *h*	−0.018 [−0.020, −0.015] *c*	−0.047 [−0.055, −0.037] *c*
DINO ViT-S	−0.105 [−0.107, −0.102] *l*	−0.083 [−0.085, −0.080] *l*	−0.146 [−0.155, −0.137] *l*

test-time register, and -0.040 [-0.042, -0.039] and -0.042 [-0.044, -0.040] on the first two for top-k training.

Tests of the contribution account. Norm weighting reduces the relative weight of attention attached to small transformed-value vectors. One consequence was registered (R1); additional checks were post hoc. *No practically meaningful gain in DINO (R1).* In DINO ViT-S, with zero high-norm attention at layers 9–12 on ImageNet-S, every difference from received attention is at most 0.004 in size (Table B.2). *More high-norm attention, more gain (post hoc).* Across images, gains have a weak positive association with the share of received attention on high-norm tokens: Spearman $+0.075$ [0.057, 0.092] and $+0.132$ [0.113, 0.150] on enrichment (DeiT-S, DeiT-B), $+0.045$ [0.032, 0.060] and $+0.158$ [0.143, 0.173] on ClickMe. *Assigned versus shuffled weights (post hoc).* Against the same weights shuffled among the patches of each head, contribution weighting is higher in all four supervised cells: $+0.038$ [0.037, 0.040] and $+0.073$ [0.071, 0.076] on enrichment, $+0.035$ [0.035, 0.036] and $+0.067$ [0.065, 0.068] on ClickMe. The shuffle control averages seeds 0–4 within each image before bootstrap resampling. Finally, also post hoc, weighting by $\|v_j^h\|$ instead of $\|W_O^h v_j^h\|$ is comparable in every cell (differences -0.010 to 0.000). This comparable verdict does not establish equivalence. C3 aggregates per-head terms related to Kobayashi et al. [19] and shares a norm-weighted component with Zhu et al. [33]; their full analyses and aggregation differ.

B.1 Parameter-randomisation sanity check

The original study included a model-parameter randomisation check [2] on 1,000 fixed-seed ImageNet-S images. Parameters were cumulatively re-initialised from the classifier and final norm through blocks 12–1 to the patch/position embeddings and CLS token. The trained model's probes and mean maps remained fixed. Received attention and its corrections are class-agnostic and therefore do not depend on classifier parameters alone. Table B.3 reports the stored median per-image Spearman correlations between trained and fully randomised maps, with layers 9–12 summed. All five readings meet the registered median < 0.5 criterion; the model-independent centre Gaussian fails as designed. This is a necessary sensitivity check, not proof of prediction faithfulness. The existing summary is reproduced here without rerunning the experiment.

Table B.3: Stored model-randomisation results: median map correlation after full randomisation. The five readings pass the registered < 0.5 criterion in every model.

Reading	DeiT-S	DeiT-B	DINO ViT-S
Received attention	−0.033	0.055	0.071
Mean-map subtraction	0.187	0.366	0.100
Probe relocation	0.063	0.004	−0.002
Contribution weighting	−0.064	0.070	0.114
All three corrections	0.069	0.078	0.025
Centre Gaussian	1.000	1.000	1.000

C Why does probe relocation fail?

Relocation lowers every deep-layer cell (Table 1). Two planned comparisons narrow the problem. Early relocation remains comparable under the registered rule (R1): at layers 1–4 it changes enrichment and ClickMe alignment by at most 0.004 in every model. And in DeiT-S and DINO ViT-S it is lower than a Gaussian matched in mean squared token displacement (Table B.2). The rest of this section is post hoc: Table C.1 lists four explanations that we tested after seeing this result; none meets its specified criterion. Most probability lies near the true patch, and replacing the linear probe with the tested MLP does not restore map quality. This does not exclude all probe-capacity or calibration limitations. The probe targets index decodability rather than input-content provenance; whether this mismatch explains the failed relocation remains untested (Section 4).

Table C.1: Four explanations for probe-relocation failure (post hoc). Model order: DeiT-S, DeiT-B, DINO ViT-S. Cells are the two centre-controlled metrics in the three models; criteria and probe settings are defined in this section.

Explanation	Test (layers 9–12 unless stated)	Result
The probe sends score to distant patches	Share of the probe's error mass more than three patches from the true patch	Small: 2–29% across models and layers. At layer 12, 45%, 66% and 81% of the probability lies in the 3×3 neighbourhood of the true patch
The probe has a fixed position preference	Peak of the probe's mean prediction relative to uniform; its correlation with the mean map	Close to uniform (peak 1.1–1.3 times the uniform value); at most weakly related to the mean map
The linear probe is too weak	Two-layer MLP probe in place of the linear one	Layer-12 accuracy rises by only 1.8, 3.8 and 6.7 points; relocation is still lower than received attention in all six cells
Global tokens are harder to place	Probe accuracy and entropy against a token's similarity to the image's mean token (layers 10–12)	No consistent relation; accuracy falls with globality only in DeiT-B

Probe capacity and explanatory criteria. The tested MLP has a 1,024-unit hidden layer, GELU and a 196-way output. It uses the linear probe's optimiser, epochs, minibatches and standardisation, trained on 1,800 of the original probe-training images with 200 for best-validation-epoch selection; the original 2,000 test images remain separate. Pre-specified post-hoc criteria were: error mass beyond three patches greater than 0.5 at every deep model/layer; fixed preference peak at least twice uniform and mean-map correlation at least 0.5 (with positive lower interval bound) in every deep model/layer; MLP layer-12 accuracy gain above 10 percentage points in some model, with relocation rescued if it is not lower in at least 4/6 alignment comparisons. These tested criteria fail; the finite MLP check cannot exclude all capacity or calibration limitations.

Globality analysis. Excluding high-norm tokens, globality is a token's cosine similarity to the mean of the other retained tokens in the image. Tokens are split into five equal-size rank groups. We relate the group rank to index accuracy and probe entropy, overall and within four equal-count radial groups, at layers 10–12. The post-hoc criterion required negative accuracy and positive entropy associations (overall intervals and within-ring means) in at least two models. No consistent relationship meets this criterion.

D Example maps

The stored gallery holds 24 ClickMe images drawn at random with a fixed seed. Fig. D.1 shows the two with the smallest ClickMe index, and Fig. D.2 the next four. The image-selection rule is unchanged. Received attention and contribution weighting look much alike, while probe relocation spreads the score over many more patches. The gains in Table 1 are averages of small changes in how each map ranks its patches, which single maps do not show.

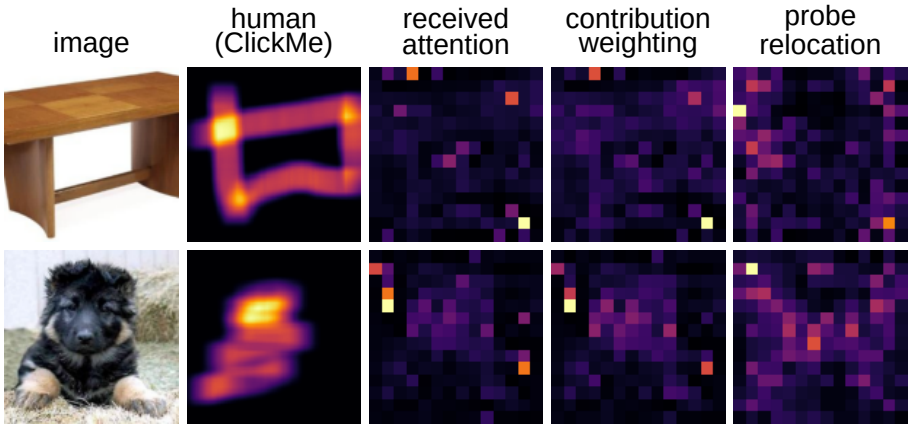


Figure D.1: The first two fixed-seed examples, DeiT-S, layers 9–12. Colours show scores min–max normalised within each map; amplitudes are not comparable across panels. These images were not selected for correction gain.

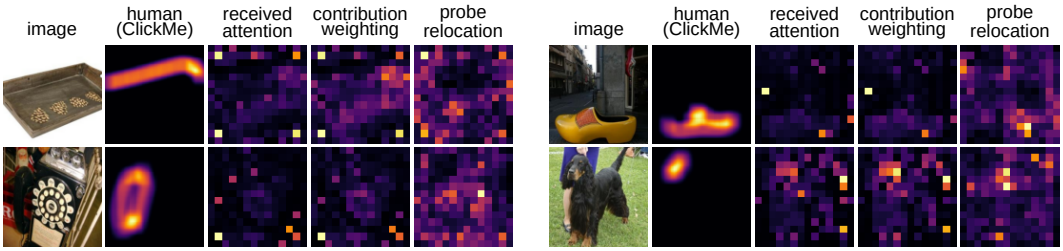


Figure D.2: DeiT-S on four further ClickMe images, layers 9–12; colours are normalised within each map and do not compare amplitudes across panels.

E Key parts of the code

Given the attention a_{ij}^h and values v_j^h of a layer, the readings are:

```
def received_and_contribution(p, v, w_o, prefix=1):
    # p: [B,H,T,T]; v: [B,H,T,d]; w_o: [D,H,d]
    # T includes CLS; values include the value-projection bias.
    gram = torch.einsum('dhe,dhf->hef', w_o, w_o)
    length = torch.einsum('bhne,hef,bhnf->bhn', v, gram, v)
    length = length.clamp_min(0).sqrt()
    received = p.sum(2)  # sum over queries
    s = received.sum(1)[:, prefix:]
    c = (received * length).sum(1)[:, prefix:]
    return s, c

def relocate(s, probs):  # source j to predicted position p
    return torch.einsum('bj,bjp->bp', s, probs)
```